

LeadPulse AI ⚡

Autonomous B2B Lead Discovery, Puppeteer + Cheerio Web Scraping & 5-Dimension AI Lead Scoring Architecture

PUPPETEER + CHEERIO ACTIVE

5-DIMENSION AI SCORER

GROQ LLAMA-3.3-70B POWERED

1. Executive Architecture Overview

LeadPulse AI is an autonomous outbound sales prospecting platform. It continually crawls live web channels to identify target technology companies matching a user's Ideal Customer Profile (ICP), extracts structured company entities, enriches executive contact emails, and evaluates prospects using a 5-dimension AI match scoring engine.

Step 1: ICP Search Query Construction

Converts target customer criteria (industries, geographies, keywords, job titles) into search queries for search engines, news feeds, job boards, and social forums.

Step 2: Multi-Source Web Scraping (Puppeteer + Cheerio)

Launches headless Chromium sessions via Puppeteer to navigate search targets and extracts titles, URLs, and snippets using Cheerio DOM and XML parsers.

Step 3: AI Entity Extraction (Groq Llama-3.3-70B)

Passes batch text chunks to Groq LLM to extract company names, website domains, industry categories, locations, and recommended decision-maker roles.

Step 4: SHA-256 Hash Deduplication

Computes SHA-256 hashes (`company_name + source`) to deduplicate prospects across all scraping channels before database insertion.

Step 5: Executive Contact Enrichment

Enriches companies with verified decision-maker emails, titles, and LinkedIn profiles via Apollo.io, Hunter.io, and Contact Synthesizer.

Step 6: 5-Dimension AI Scoring & Groq Score Insights

Calculates 0–100 match scores based on Intent (30%), Profile Fit (25%), Company Fit (20%), Recency (15%), and Engagement (10%), with Groq LLM score cards.

2. Live Web Scraping Engines (Puppeteer + Cheerio)

LeadPulse AI utilizes four dedicated discovery modules designed for high-concurrency web crawling:

Scraper Engine	Target Web Channels	Scraping Methodology
Deep Web Scraper	DuckDuckGo Search, Technology Catalogs, Web DOMs	Puppeteer headless browser navigation + Cheerio DOM parsing (<code>.result__title</code> , <code>.result__snippet</code> , <code>.result__url</code>) + Axios fallback
News & PR Scraper	Google News RSS, Press Release Feeds	Fetches RSS XML payloads and parses nodes using Cheerio XML mode (<code>cheerio.load(data, { xmlMode: true })</code>)
Job Intent Scraper	RemoteOK, Jobicy API, HackerNews Who's Hiring	Monitors live tech hiring posts to detect engineering budget allocation and active expansion signals
Social Signals Scraper	Reddit Subreddits (r/devops, r/SaaS, r/sysadmin)	Crawls public pain-point discussions using Reddit JSON API & RSS feeds with Cheerio parsing

3. 5-Dimension AI Scoring Engine & Weights

Every prospect is evaluated through a five-dimension mathematical scoring matrix:

$$\text{Total Score} = (\text{Intent} \times 0.30) + (\text{Profile Fit} \times 0.25) + (\text{Company Fit} \times 0.20) + (\text{Recency} \times 0.15) + (\text{Engagement} \times 0.10)$$

Dimension	Weight	Evaluation Criteria
Intent Signals	30%	Hiring posts (+35), news mentions (+25), social mentions (+20), web mentions (+20)
Profile Fit	25%	Contact title match against target ICP (Exact +95, Partial +80, Seniority +70)
Company Fit	20%	Industry match (+35), company size match (+25), geography (+20), website (+5)
Recency Decay	15%	Exponential signal decay based on discovery date (14-day half-life)
Engagement	10%	Pipeline status progression (<code>accepted</code> +80, <code>reviewed</code> +50, <code>scored</code> +20)

Scoring Tier Mapping

- 🔥 **Hot Tier (70–100):** High conversion probability with strong hiring intent and decision-maker match.
- 🌤️ **Warm Tier (40–69):** Good profile match with moderate intent signals.
- ❄️ **Cold Tier (0–39):** Low signal frequency or outdated discovery timestamp.

4. Groq LLM Score Explanation Briefing Cards

For every scored prospect, the system passes lead data and score breakdowns to **Groq AI (Llama-3.3-70B)** to produce a concise 2–3 sentence briefing card for sales representatives.